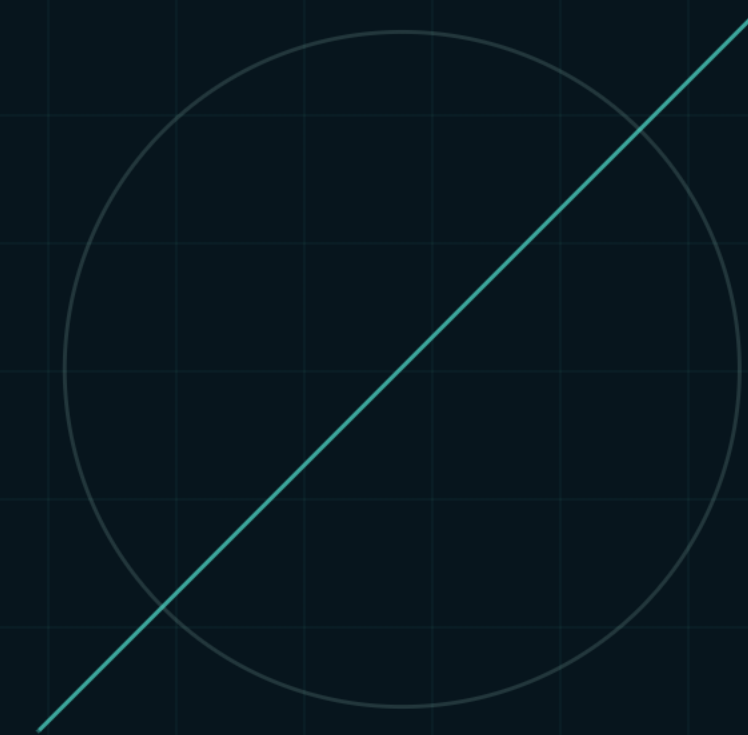


01 / EMBEDDED SYSTEMS

ESP32 Hardware



| A tiny board with the reach of a whole lab.

ACTUATOR 26

FIELD NOTES / 2026

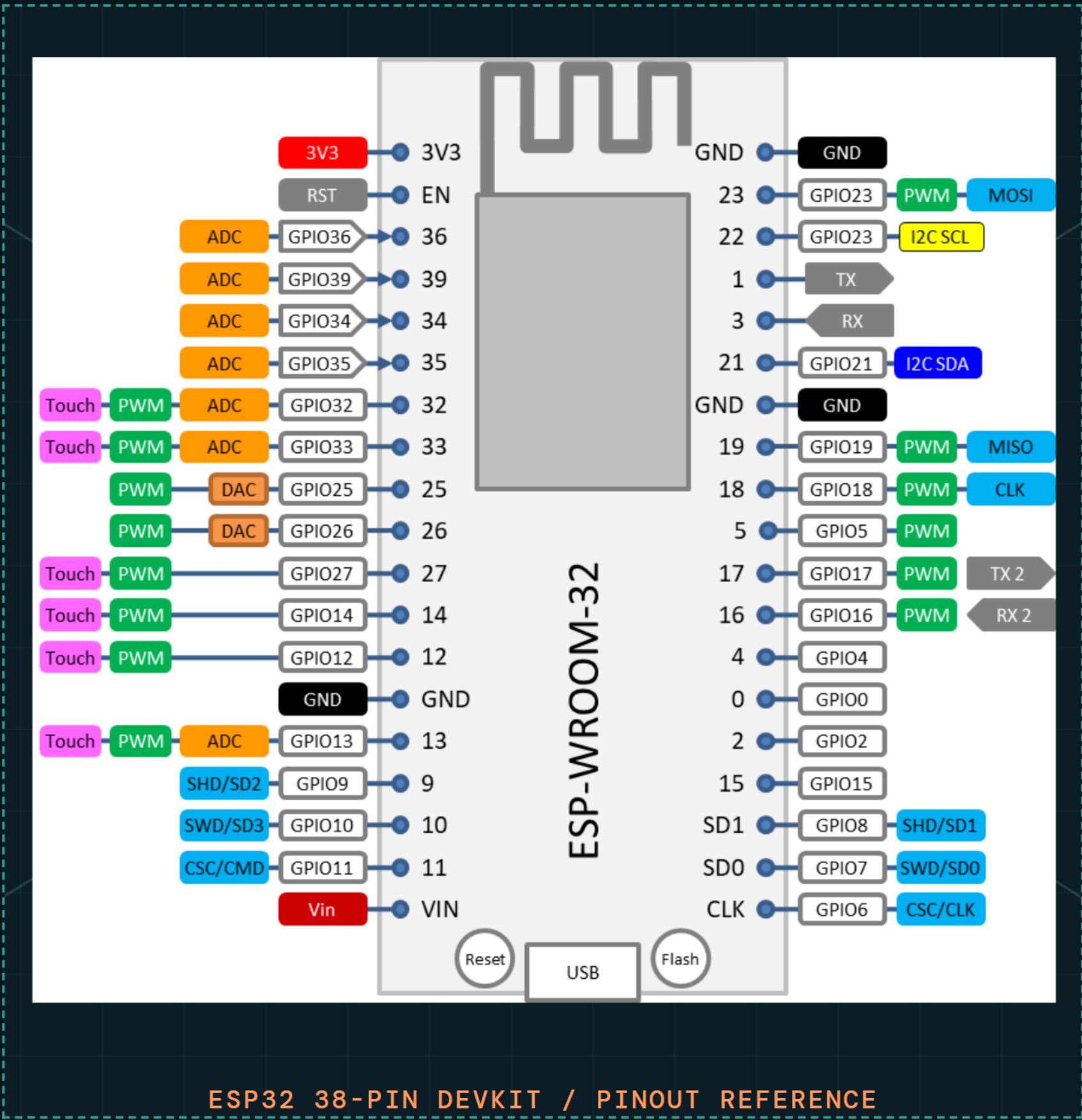
02 / IDENTIFY THE BOARD

ESP32

38-pin breakout.

A development board built around the ESP32-WROOM-32 module, exposing 19 header positions on each side.

38	3.3V	240
HEADER	GPIO LOGIC	MHZ DUAL
PINS		CORE



03 / THE 38-PIN LAYOUT

Read the headers from the outside in.

Pin names vary slightly between manufacturers. Always compare the silkscreen on your board with its datasheet before wiring.



*IC style reference. Header order and labels can vary by manufacturer; verify the silkscreen before wiring.

04 / COMPLETE SIGNAL REFERENCE

What each group is good for.

POWER

VIN · 3V3 · GND

VIN accepts regulated 5 V through the board regulator. 3V3 is the regulated output. Connect grounds together before testing.

DIGITAL I/O

GPIO 0–33

General-purpose digital pins. Most support input, output, interrupts, and PWM. GPIO 34–39 are input-only.

ANALOG

ADC1: 32–39

Use ADC1 for analog sensors when Wi-Fi is active. ADC2 conflicts with Wi-Fi and is less dependable for that use.

SPECIAL

DAC · TOUCH · RTC

DAC output: GPIO25/26. Capacitive touch: GPIO0, 2, 4, 12–15, 27, 32, 33. RTC GPIOs support low-power wake.

05 / COMMUNICATION DEFAULTS

One board, many buses.

BUS	DEFAULT PINS	TYPICAL HARDWARE	NOTES
I ² C	GPIO 21 SDA GPIO 22 SCL	OLED, IMU, sensors	Pull-up resistors are normally required.
VSPI	GPIO 23 MOSI 19 MISO · 18 SCLK · 5 CS	SD card, displays	Chip select is software-controlled.
UART2	GPIO 17 TX2 GPIO 16 RX2	GPS, motor controllers	Use a common ground and matching logic levels.
UART0	GPIO 1 TX0 GPIO 3 RX0	USB serial monitor	Shared with programming and debug.

06 / PIN MAP AT A GLANCE

A quick-reference cheat sheet.

SAFE STARTERS

GPIO 16, 17, 18, 19, 21, 22, 23, 25, 26, 27, 32, 33
Good general choices after checking your peripheral requirements.

FLASH RESERVED

GPIO 6–11
Connected to the module’s SPI flash. Do not use them as breakout pins.

INPUT ONLY

GPIO 34, 35, 36 / VP, 39 / VN
Read sensors here. They cannot drive an LED, relay, or motor.

BOOT SENSITIVE

GPIO 0, 2, 5, 12, 15
External pull-ups or pull-downs can stop the board from booting.

07 / POWER AND PROTECTION

Three rules before power-up.

-  **Use 3.3 V logic**
ESP32 GPIO is not 5 V tolerant. Level-shift 5 V sensor signals.
-  **Share the ground**
Every external module needs a common GND reference with the ESP32.
-  **Feed motors separately**
Motor noise and current spikes belong on their own supply, with a driver between.

CHECK THIS FIRST

Never connect 5 V directly to a GPIO.

Power the board through USB, VIN, or a clean regulated 5 V source. Confirm polarity and current capacity before attaching a battery.

5 V  3.3 V  GPIO

08 / FROM CODE TO MOTION

Sense. Decide. Act.

A reliable embedded project keeps the loop simple: collect a signal, make a decision, produce a visible result.

GPIO → peripheral → logic → output

```
// read, decide, respond
const distance = readSensor();
if (distance < SAFE_LIMIT) {
    stopMotors();
    signalWarning();
} else {
    driveForward();
}
```



09 / THANK YOU

Thank You.

Let's learn how to program and turn ideas into real hardware.

ESP32 · CODE · CREATE

LET'S
PROGRAM

